On-farm intervention guidelines for responding to outbreaks of avian influenza in dairy cattle

Overview

Highly pathogenic avian influenza viruses belonging to clade 2.3.4.4b have circulated widely in bird and mammal populations since 2020. Introductions of H5N1 clade 2.3.4.4b into dairy cattle from wild birds have resulted in outbreaks on dairy farms across the United States. Outbreaks have serious consequences on animal health, result in significant economic losses, and pose an immediate threat to human health through spillover infections and the contamination of raw milk products. H5N1 clade 2.3.4.4b was first detected in Australia in June 2026. AusVet has estimated the risk to the Australian dairy industry to be low [1]. However, to minimise the risk and size of outbreaks on dairy farms we have developed testing and intervention guidelines based on the results of a model-based analysis.

Methods

We used mathematical modelling informed by international data to: investigate dairy farm conditions associated with the highest outbreak risk; and assess the effectiveness of on-farm intervention strategies. The analysis assumes that the primary route of transmission is via contamination of milking apparatus by infected cattle.

Factors contributing to risk

We estimated an increased risk and size of H5N1 outbreaks on dairy farms where:

- Cattle are milked more frequently.
- There are a higher number of cattle per milking unit.
- Milking units are cleaned less frequently.

The frequency of milking had the greatest impact on the risk and size of H5N1 outbreaks.

Recommendations on risk mitigation

In the event of influenza H5N1 outbreaks affecting cattle in Australia, pre-emptive intervention measures can be taken to mitigate the risk and reduce the size of outbreaks on dairy farms.

<u>Pre-emptively separate herds into ordered milking cohorts</u>

When possible herds should be divided into distinct milking cohorts, where cohorts are kept in separate paddocks and milked in the same order each day. The greater the number of milking cohorts, the lower the risk and size of an outbreak. If herds are already kept in separate mobs, then milking the mobs in a consistent order ensures the same risk mitigation (they can be considered milking cohorts).

<u>Place imported cattle into the final milking cohort</u>

When importing new cattle onto a dairy farm they should be kept in the last milking cohort. This will limit the risk of outbreaks in other milking cohorts. The last milking cohort could also be considered as a quarantine cohort (i.e. newly imported cattle should be kept separate from the rest of the herd and milked last). This is in line with guidance from AusVet that suggests newly imported cattle should be quarantined for 14 days to reduce the risk of outbreaks [1].

<u>Regular and strategically timed cleaning of milking units</u>

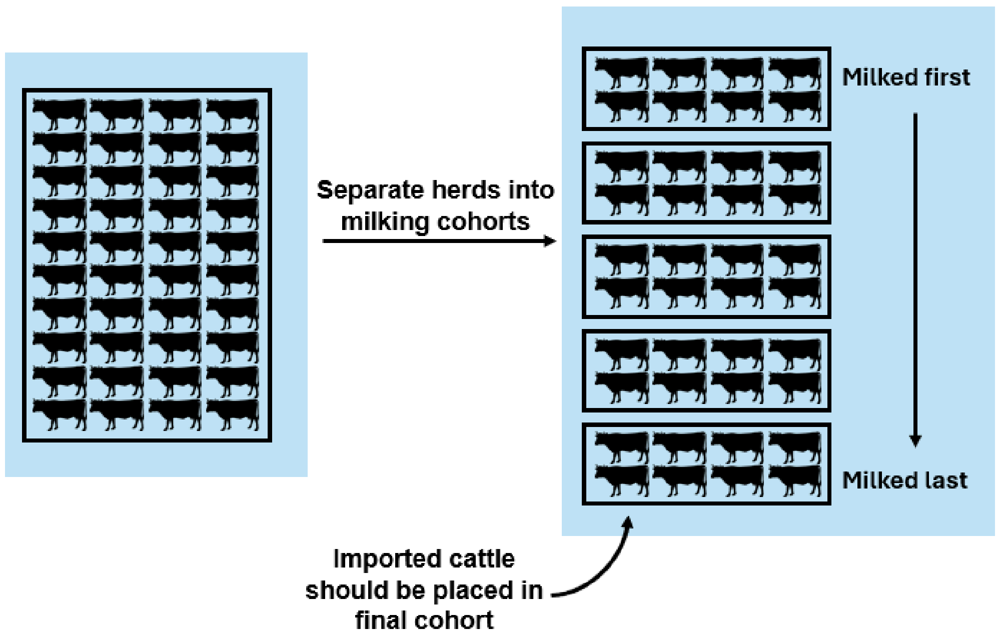
Milking units should be cleaned regularly, and in particular between milking of milking cohorts. When there are constraints to how often milking units can be cleaned, the most effective timing of cleaning depends strongly on the duration of milking unit infectiousness, which is currently poorly understood. Where cleaning is highly effective (i.e. complete decontamination), cleaning between milking of middle cohorts is best (e.g. between cohorts 2 and 3 if 5 cohorts total).

<u>Reactively separating herds into ordered milking cohorts when an outbreak is detected</u>

It may be difficult for dairy farms to separate herds into milking cohorts for a sustained period. Reactively separating herds into milking cohorts (or increasing the number of milking cohorts) when an outbreak is detected can still be effective if the outbreak is detected early.

<u>Frequently testing bulk milk samples</u>

RT-PCR testing of bulk milk samples from the central milk vat will be highly sensitive at detecting an outbreak, even when there are few infected cattle on a farm. Frequently testing bulk milk samples (e.g. weekly) will allow outbreaks to be rapidly detected ensuring reactive measures can be effective. Bulk milk sample testing could also be used before the transport of cattle between dairy farms.



Limitations

There is a high degree of uncertainty around the biological and farming factors influencing transmission. The primary route of transmission between dairy cattle is considered to be through the repeated use and contamination of milking apparatus by infected animals. However, it is also possible that a degree of transmission occurs via contamination of the environment (e.g. floor and bedding or bails) with an infectious animal's milk, or direct respiratory transmission between cattle.

The intervention recommendations presented here are all expected to have a beneficial effect, reducing the risk and size of outbreaks. However, the magnitude of the effect will depend on the transmission regime and will vary between dairy farms.

Further guidance is available from the Department of Agriculture, Fisheries and Forestry:

1. Risk assessment of H5 bird flu in Australian dairy cattle and other livestock. Available: https://www.agriculture.gov.au/biosecurity-trade/pests-diseases-weeds/animal/avian-influenza/livestock